

Perfect Circle — Requirement Document

Prepared by: Manya Bansal · Sep 27, 2026 · Version 0.1

Perfect Circle is a free web game where players draw a shape freehand around a dot and get an accuracy score. Version 0 (v0) is the first working release: it adds four shapes, four game modes, 1v1 duels, a daily challenge and progression, and runs on the author's own PC as a local server that phones, tablets and laptops on the same Wi-Fi can open.

Purpose and scope

- In scope: circle, square, triangle and star drawing; time limit, disappearing ink, moving dot and non-dominant hand modes; online duels on two devices and split-screen duels on one device; a daily challenge with a shareable result; streaks, badges and a history graph; running on a personal PC as a local server (localhost) that other devices on the same Wi-Fi can open.
- Out of scope for v0: public internet hosting, a domain name, Play Store and App Store builds, payments or ads, in-game chat, friend lists.

Stakeholders

Stakeholder	Interest
Manya Bansal (product owner and author)	Defines scope, approves releases, signs off testing
Casual players (phone)	Quick rounds, sharing results, beating friends
Students and teachers (laptop, smartboard)	Classroom play on a shared screen, split-screen duels
Development team	Clear requirements that can be built and tested

Functional requirements

Priority uses MoSCoW: Must, Should, Could.

ID	Requirement	Priority
FR-01	The player can choose a shape: circle, square, triangle or star.	Must

ID	Requirement	Priority
FR-02	The game shows a centre dot and a faint hint of the chosen shape before drawing starts.	Must
FR-03	The game scores each attempt from 0 to 100% with one decimal, based on how closely the stroke matches the ideal shape.	Must
FR-04	The stroke is coloured per segment from green (accurate) to red (off), and the ideal shape is overlaid after the attempt.	Must
FR-05	The game rejects an attempt that is too close to the dot, goes backwards, or does not close the shape, and says why.	Must
FR-06	Time limit mode: the attempt must finish within a countdown of 5, 3 or 2 seconds.	Must
FR-07	Disappearing ink mode: the stroke fades out 0.4 seconds after it is drawn.	Must
FR-08	Moving dot mode: the centre dot moves slowly on a fixed path and the score is measured against its live position.	Should
FR-09	Non-dominant hand mode: the player declares they are using their other hand; results are tagged and tracked separately.	Should
FR-10	A player can create a duel room and get a 4-character code and a QR code.	Must
FR-11	A second player can join the room from another device with the code or QR.	Must
FR-12	Both players get the same shape and mode, a synced 3-2-1 countdown, and draw at the same time; best of 3 rounds wins.	Must
FR-13	Split-screen duel: on one tablet or touch laptop, the screen splits in two and both players draw at once.	Should
FR-14	On a single-mouse laptop, split-screen duel runs turn by turn.	Could
FR-15	Every day has one daily challenge (shape and mode), the same for every player on the server, resetting at 00:00 UTC.	Must

ID	Requirement	Priority
FR-16	A player gets one scored attempt at the daily challenge per day.	Must
FR-17	After the daily attempt, the player can share a text result (challenge number, score, coloured squares) like Wordle.	Must
FR-18	The daily challenge shows a top-100 leaderboard of players on the server, with nicknames.	Could
FR-19	The game counts a daily streak: consecutive days with a completed daily challenge.	Must
FR-20	The game awards badges, for example First 90%, First 95%, 7-day streak, All shapes above 85%, First duel win.	Must
FR-21	A history graph shows the player's scores over time, filterable by shape and mode, for 7, 30 or 90 days.	Must
FR-22	The player can play without an account; progress is saved on the device.	Must
FR-23	The player can log in with a nickname and a 4-digit PIN to see the same progress on phone and laptop.	Should
FR-24	The player can set a nickname shown in duels and leaderboards.	Should
FR-25	The game works with mouse, touch and stylus.	Must
FR-26	The game can be installed as an app window on the host PC; phones and other laptops can install it when the server runs on local HTTPS.	Must
FR-27	Solo modes work offline once the app has been loaded.	Should
FR-28	The game supports light and dark themes.	Could
FR-29	The whole game runs on a personal PC as a local server; one command starts it at <code>http://localhost:3000</code> .	Must
FR-30	Phones and laptops on the same Wi-Fi join by opening the PC's local IP address, for example <code>http://192.168.1.5:3000</code> .	Must

Non-functional requirements

ID	Area	Requirement
NFR-01	Performance	Drawing renders at 60 frames per second on a mid-range phone (for example Redmi Note series).
NFR-02	Performance	First load under 2 seconds over the local Wi-Fi; app bundle under 300 KB gzipped.
NFR-03	Performance	Duel score updates reach the other player in under 200 ms on the local network.
NFR-04	Compatibility	Latest 2 versions of Chrome, Safari (iOS 16+), Edge and Firefox; screens 360 px to 2560 px wide.
NFR-05	Accessibility	Menus usable by keyboard; score shown as a number, not colour alone; text contrast meets WCAG AA.
NFR-06	Security	Scores in duels and the daily leaderboard are recalculated on the local server from the submitted stroke.
NFR-07	Privacy	Only a nickname and a hashed PIN are stored, on the host PC only; no data leaves the local network. Complies with India's DPDP Act 2023.
NFR-08	Availability	Available whenever the host PC is on and the server is running; data survives restarts.
NFR-09	Cost	No hosting cost; runs on a Windows, macOS or Linux PC with Node.js 20 and 4 GB RAM, serving up to 30 players at once.
NFR-10	Maintainability	Scoring logic covered by unit tests at 90% or more line coverage.

Assumptions and constraints

- v0 is a pre-release for personal and classroom use; public hosting can come in a later version.
- The host PC and all players' devices are on the same Wi-Fi network, and the PC's firewall allows port 3000.

- Players can reach the game only while the host PC is on and the server is running.
- Non-dominant hand use cannot be verified; it relies on the player's honesty.
- Online duels need both devices on the same network as the host PC; split-screen duels need only one device.
- The daily challenge uses UTC so every player sees the same shape on the same day.

Open questions

- Should the daily leaderboard (FR-18) be in v0 or a later v0.x release?
- Should the server use local HTTPS so phones can install the app (FR-26), or is playing in the browser enough for v0?